(23) What is rank of a word 'GOOGLE' if we arrange in dictionary alphabetical order?

Ans: first arrange letters in 'google' in alphabetical order.

google → e g g l o o [repetition allowed]

e _____ → $\frac{5!}{2! \cdot 2!} = \frac{120}{4} = 30$

g e ____ → $4!/2! = \frac{24}{2} = 12$

g g ____ → $4!/2! = \frac{24}{2} = 12$

g l ____ → $4!/2! = 24/2 = 12$

when Repetition is allowed

g o e ___ → $3! \longrightarrow 6$

g o g ___ → $3! \longrightarrow 6$

g o l ___ → $3! \longrightarrow 6$

g o o e __ → $2! \longrightarrow 2$

g o o g e _ → $1! \longrightarrow 1$

g o o g l e → $1 \longrightarrow 1$

88

S.c: e g g l o o

$1\left(\frac{5!}{2!\,2!}\right) + 3\left(\frac{4!}{2!}\right) + 3(3!) + 1(2!) + 1(1!) + 1 = 88$

(24) Rank of 'PASSION' when we arrange in dictionary alphabetical order?

Ans: PASSION → I N O P S S

A _____ → $\frac{6!}{2!} = \frac{720}{2} = 360$

I _____ → $6!/2! = 360$

N _____ → $6!/2! = 360$

O _____ → $6!/2! = 360$

P A I ____ → $4!/2! = 24/2 = 12$

P A N ____ → $4!/2! = 12$

P A O ____ → $4!/2! = 12$

(24)

PASI ___ = 3! = 6
PASN ___ = 3! = 6
PASO ___ = 3! = 6
PASSIN_ = 1! = 1
PASSION = 1 = 1

Rank = 4(360) + 3(12) + 3(6) + 1 + 1 = 1440 + 36 + 18 + 2
= 1496

(25) If we arrange letters of the word 'TABLE' in alphabetical order what is 45th word?

TABLE → A (B) E L (T)

A _____ → 4! = 24 words completed

B _____ → 4! = 24 words completed

BA ___	→ 3! = 6	48 words completed but we want 45th so, it starts with 'B' only then continue with 'B' only
BE ___	→ 3! = 6	
BL ___	→ 3! = 6	
BT ___	→ 3! = 6	
	24	

24 + 24 = 48 but we want 45th so, it starts with 'BT'... continue with 'BT'

BTA __ → 2! = 2 [24 + 18 + 2 = 44 words Completed]

Next word only our answer.

BTEAL → 1 → 44 + 1 = 45th word.

BTELA

BTEAL → 45th word

(26) If we arrange letters of the word 'MOTHER' in alphabetical order, what is 250th word?

Ans: MOTHER → EHMORT.

MOTHER ⇒ Total words = 6! = 720

E _ _ _ _ _ = 5! = 120
H _ _ _ _ _ = 5! = 120
240

(So, it starts with 'M' only)

M E H _ _ _ = 3! = 6
M E O H _ _ = 2! = 2
M E O R H T ⟶ 1
M E O R T H ⟶ 1
250

ME _ _ _ _ then 240+24 > 250.

So, ME.

ME O _ _ _ = 3! = 6
240+6+6 > 250

So starts with ME O

250th word is 'MEORTH'.

(27) From a word 'HANDSOME' how many 5 lettered words can be formed such that 2 vowels & 3 consonants should be there?

Ans:-

HANDSOME.

vowels → [AOE], Consonants → [D H M N S]

here in this question we have to select (combination) 2 vowels from 3 vowels & 3 consonants from 5 consonants & arrange (permutation) them to form 5 lettered word.

↳ Combination (selection):- $^3C_2$ (vowels) × $^5C_3$ (consonants)
= 3 × 10 = 30 ways selection.

↳ Permutation (arrangement):- Selected 5 letters in 5 places to form 5 lettered word. in 5! ways. = 120 ways

↳ Selection & Arrangement:- 30 × 120

~~[Permutation]~~ [Combination] × [Permutation]

= 3600 words can be formed.